cowl
The cover—often removable or partly removable—surrounding an aircraft's engine. *See also* cowl gills.

cowl gills
Perforations in the cowls of some aircraft engines. They can be covered or uncovered, and when open let through a flow of air that cools the engine. *See also* cowl.

crankcase
The part of a piston engine that houses the crankshaft. *See also* crankshaft.

crankshaft
The shaft in a piston engine that converts the reciprocating (to-and-fro) motion of the pistons into the rotational motion needed to turn a propeller.

critical altitude
The maximum altitude at which an aircraft engine aided by a supercharger can maintain full power. Above this altitude, the engine power drops as air density—and thus available oxygen—reduces.

crosswind
A wind blowing sideways across the runway in use, making takeoffs and landings more difficult, and sometimes preventing them altogether.

cyclic control
A control in a helicopter that changes the angle of the rotating blades. Adjustments to this control enable the helicopter's speed and direction to be altered. *See also* collective control.

cylinder
A combustion chamber in a piston engine. *See also* cylinder block, piston engine.

cylinder block
A block of (usually cast) metal into which cylinders (combustion chambers) have been bored. This forms the main body of a piston engine. *See also* cylinder, piston engine.

cylinder head
The upper part of a piston engine that is attached to the top of the cylinder block. It houses the combustion chambers and spark plugs, as well as valves that regulate gas entry and exit.

delta wings
A wing arrangement in which the leading edge of each wing is swept back diagonally in a straight line, but the trailing edge is joined perpendicularly to the fuselage. The overall wing silhouette is like a triangle or a Greek letter Delta.

designation
In aviation contexts, a method of classifying aircraft, especially in the United States Air Force, and also the classification of individual aircraft within this system.

diffuser
Any air pipe or duct, such as a section of a jet engine, that gets wider in the downstream direction. This widening has the effect of slowing down gases passing through it.

dihedral
An arrangement of an aircraft's wings in which they are angled upward from the horizontal, so that the tips of the wings are higher than the point at which they are attached to the fuselage. *See also* anhedral.

dirigible
A more technical term for an airship, meaning "steerable" or "directable," in contrast to a traditional balloon.

dogfight
A battle at close quarters between groups of opposing military aircraft. Mainly a feature of past wars, when aircraft were slower, the word "dogfight" implies a series of individual fights rather than an attempt to attack in formation.

drag
The force (air resistance) that impedes the motion of an aircraft. There are different kinds of drag, including drag created by air friction against an aircraft's skin, and drag caused by turbulent air around an aircraft.

drag coefficient
A measure of the tendency of an object to create drag when air flows past it. The higher the drag coefficient, the greater the tendency. Objects with the same surface area but different shapes can have quite different drag coefficients. *See also* drag.

drift indicator
An instrument indicating an aircraft's angle of "drift"—the difference between an aircraft's projected flight path and its actual heading, as affected by winds.

drone
An informal name for an unmanned aerial vehicle. *See also* unmanned aerial vehicle.

ejection seat
A special seat in a military aircraft that uses a rocket motor to blast the pilot clear of the aircraft in an emergency, and parachute him/her to safety.

electronic flight instrument system (EFIS)
The arrangement that now predominates in modern aircraft cockpits, in which instrument displays are fully electronic, in the form of flat screens, rather than featuring dials with mechanical pointers.

elevator
An aircraft control surface that affects the pitch of an aircraft (whether it is going up, down, or is level). The elevators are usually on the tailplane of an airplane. *See also* control surface, pitch, tailplane.

elevon
An aircraft control surface that functions both as an aileron and as an elevator. *See also* aileron, elevator.

empennage
A tail assembly—the overall arrangement of an airplane's tail, including (in most aircraft) a tailplane, tail fin, rudder, and elevators. The term comes from a French word meaning "to feather an arrow."

exhaust manifold
A structure attached to a piston engine that collects the exhaust gases coming from several different cylinders and directs them into a single exhaust pipe.

exhaust port
A pipe that carries away exhaust gases from a cylinder of a piston engine.

fairing
Any streamlined covering added to part of an aircraft.

Federal Aviation Administration (FAA)
The public authority in the USA that regulates aviation, including air traffic control. *See also* Civil Aviation Authority

ferry tank
An additional fuel tank carried by an aircraft to extend its range.

firewall
An internal wall or barrier used to resist the spread of fire.

flaps
Control surfaces on the trailing edge of a wing that tilt downward and/or extend backward to increase the wing's lift. Their main use is in takeoff and landing. There are various kinds of flaps, some of which are no longer used today, including blown, drag, Fowler, lift, plain, slotted, and split flaps. In slotted flaps, small openings (slots) remain between the back edge of the main wing and the flap, allowing high-pressure air from below the wing to flow through to the upper surface, creating additional lift. *See also* aileron.

flight
In the sense of "a flight," the term refers to a military unit within an airforce, smaller than a squadron.

flight engineer
As airplane engines and equipment became more complex during the 20th century, this dedicated professional was necessary on many larger aircraft to supervise technical aspects of the equipment during flight. With increasingly sophisticated computer controls, they are no longer needed on most aircraft.

flight level
A standardized measure of a plane's altitude that takes into account variations in air pressure, which can affect traditional altimeter measurements. Its significance lies in determining that the relative heights of aircraft are known accurately, avoiding any possibility of collision.

flutter
The undesirable vibrating of a wing or other part of an aircraft.

fly-by-wire
An electronic flight control system used instead of mechanical controls.

four-stroke engine
A type of piston engine in which the cycle of operation—introducing the fuel to the cylinder, burning it, and getting rid of exhaust gases—requires four piston strokes. *See also* piston engine, two-stroke engine.

fuel injection
A system in which fuel is dispersed into tiny droplets and actively pumped into the manifold combustion chamber of an engine. It has largely replaced the carburettor. *See also* carburettor.

fuselage
The main body of an airplane, not including the wings and tail. The word comes from the French term for spindle, as many early fuselages were roughly spindle-shaped.

g-force
The force experienced by the crew of an aircraft undergoing rapid acceleration or deceleration. One "g" is equal to the force usually exerted by Earth's gravity.

gas turbine
An engine that operates via turbine wheels extracting energy from hot burned gases, and converting this energy into rotary motion. All jet engines except ramjets incorporate gas turbines.

general aviation (GA)
A term for all aviation except for military aviation and scheduled commercial flights. For example, recreational aviation counts as general aviation. There are many more airfields servicing general aviation than there are airports for scheduled flights.

glass cockpit
A type of cockpit in which the instrument displays involve digital electronic screens and head-up displays rather than traditional mechanical dials.

ground effect
The altered aerodynamics experienced by an airplane when flying close to the ground, involving both increased lift and decreased drag.

gull wing
An aircraft wing that has a sharp bend along its length, like the wing of a seagull.

hang glider
A glider in which the human flyer hangs in a harness below the wing, controlling and steering largely through changes in body position. Although some early-built gliders such as those made by Otto Lilienthal in the 19th century could be classed as hang gliders, the term itself was not used until the 20th century.

head-up display (HUD)
A unit that projects information, such as combat status and aircraft performance data, onto a transparent screen in the pilot's line of sight, lessening the need to look down into the cockpit.

horsepower
A traditional unit of measurement of the power of an engine, generally taken as equivalent to 746 Watts.

hydraulics
In aircraft, this term usually refers to systems of hydraulic power, in which fluids are forced through pipes under pressure to move wing control surfaces and undercarriages, for example.

hypersonic
A flight speed that excels Mach 5—five times the speed of sound.

impeller
A bladed wheel in some jet engines that compresses incoming air by flinging it out in a centrifugal direction as it passes through. *See also* compressor.

inertial navigation system (INS)
A cockpit device that provides positional and navigational information without the need for data from external references.